ODD+D Appendix for Searching Legislative Design Space

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This appendix documents the congressional institutional simulator using the ODD and ODD+D model-description conventions [Grimm et al. 2010; Müller et al. 2013]. It describes entities, state variables, scheduling, adaptation, input data, submodels, outputs, calibration screening, assumptions, and reproducibility commands for the simulator used in *Searching Legislative Design Space for Compromise and Productivity.*

CCS Concepts: • **Computing methodologies** → **Modeling and simulation**; • **Human-centered computing** → *Collaborative and social computing theory, concepts and paradigms.*

Additional Key Words and Phrases: ODD+D, legislative simulation, institutional design, model documentation

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1 Purpose

The model compares legislative institutional designs under shared generated political worlds. Its purpose is to search for mechanisms that improve compromise and productivity while preserving representative responsiveness, public-benefit alignment, minority protection, and resistance to capture. It is not a forecast of a specific Congress. It is a mechanism-search simulator: the same generated legislators and bills are routed through many institutional processes so differences can be attributed to procedural rules and incentive structures.

2 Entities, State Variables, and Scales

2.1 Legislators

Legislators have an identifier, party label, ideology, compromise preference, party loyalty, constituency sensitivity, lobbying susceptibility, reputation sensitivity, district preference, district intensity, affected-group sensitivity, and a representation profile. The representation profile records chamber eligibility, represented population share, district magnitude, region label, elected/appointed mode, lower- and upper-house term lengths, renewal cohort, renewal cycle, next renewal round, and selector body. These values determine how the legislator weighs ideology, party pressure, constituent/public support, lobbying pressure, compromise incentives, reputation, and concentrated harm before casting a final yes-or-no vote.

2.2 Bills

Bills have an identifier, title, proposer, proposer ideology, original and current policy position, public support, generated public benefit, public-benefit uncertainty, lobby pressure, private gain, salience, issue domain, anti-lobbying-reform flag, affected group, affected-group support, concentrated harm, compensation cost, cosponsor counts, amendment movement, lobbying channel spend, challenge status, citizen-review signals, active-law signals, and alternative-selection signals.

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2.3 Lobby Groups

Lobby groups are explicit collective actors. Each group has an identifier, primary issue domain, issue-preference map, preferred policy position, budget, influence intensity, defensive multiplier, information bias, public-campaign skill, capture strategy, and public-support mismatch tolerance. Groups can spend through direct pressure, agenda-access pressure, information distortion, public campaigns, litigation or delay threats, and defensive anti-reform pressure.

2.4 Institutional Processes

An institutional process is a composable module. Processes can screen proposal access, apply leadership agenda control, modify public signals, spend lobbying or attention resources, route bills to procedural tracks, mediate amendments, evaluate affected-group harm, select among alternatives, conduct chamber votes, revise through conference committees, apply vetoes, apply judicial review, route through citizen initiatives, register enacted laws, or review laws later. The chamber-structure supplement adds chamber-specification, chamber-origin routing, second-chamber amendment and veto powers, last-offer bargaining, lower-house override, committee-assignment configuration, committee-role power, eligibility and retention filters, and ex ante review through insulated fiscal, electoral, audit, ethics, or legal-review bodies. The current simulator also includes bounded linked modules for district-population public signals, long-horizon proposer strategy, omnibus bargaining, campaign-finance influence systems, and constitutional-court architecture. The final decision still resolves to a yes-or-no outcome, enactment or rejection, and a post-bill status quo. The submitted comparison includes synthesized portfolio hybrids that combine low-risk fast passage, middle-risk pairwise alternatives and mediation, high-risk citizen and harm review, proposal bonds, anti-capture safeguards, law-registry correction, and, in the expanded variant, public-opinion, influence-system, omnibus, and court-review proxies.

3 Process Overview and Scheduling

Each run follows a fixed schedule:

(1) Generate a world containing legislators, party positions, lobby groups, a status quo, and a bill stream.
(2) Build each scenario's legislative process from that world.
(3) For each bill, create a deterministic vote context containing party positions, status quo, and seeded randomness.
(4) Pass the bill through the scenario process chain.
(5) Record the bill outcome, including agenda disposition, vote totals, enactment, amendments, challenges, lobbying signals, public-review signals, and policy movement.
(6) Update the scenario's status quo when policy is enacted or rolled back.
(7) Aggregate outcomes into scenario metrics.

Campaigns repeat this over many runs and assumption cases. Scenarios share generated worlds but use independent deterministic random streams, supporting paired comparison among institutions.

4 Design Concepts

4.1 Basic Principles

The model assumes legislative outcomes are shaped by proposal access, agenda scarcity, voting thresholds, committee and information structures, lobbying and capture, amendment opportunities, public review, concentrated harm, reversibility, and strategic actor adaptation.

4.2 Emergence

Productivity, compromise, capture, volatility, legitimacy, and gridlock emerge from bill-by-bill interactions. No actor directly optimizes the reported global metrics.

4.3 Adaptation

Proposers adapt over repeated bills. They can reduce proposal volume, delay timing, moderate risky bills, seek outside-bloc cosponsorship, reduce lobby exposure, add compensation amendments, lower risk tolerance after bad outcomes, and withdraw weak risky bills. Proposer state includes trust, proposal pace, risk tolerance, concession rate, lobby exposure, cooldown, and streak counters.

Lobby groups adapt over repeated bills. They track returns by channel, select capture strategies from learned returns and current bill conditions, change issue-specific spending multipliers, adjust overall budget intensity, increase reform-threat sensitivity when anti-lobbying bills become threatening, and maintain defensive pressure against reforms that would reduce lobbying influence.

Institutional state also adapts through proposal credits, proposal bonds, law registries, sunset review, challenge-token exhaustion, attention-token spending, norm erosion, judicial reversals, and active-law correction.

4.4 Objectives

Legislators vote according to weighted incentives. Proposers seek agenda access and enactment while learning to avoid reputational or procedural failure. Lobby groups seek policy movement toward their issue preferences and private gains while resisting anti-lobbying reforms. Institutions do not optimize a single welfare function; they impose constraints and incentives whose tradeoffs are measured afterward.

4.5 Learning and Prediction

Learning is deterministic and bounded rather than a full reinforcement-learning procedure. Actors use local outcome signals: enactment, public support, public benefit, capture risk, concentrated harm, salience, private gain, prior failures, and prior channel returns.

4.6 Stochasticity

World generation, committee composition, citizen panels, bill signals, party-system profiles, and some routing decisions use seeded randomness. Identical seeds reproduce identical outputs.

4.7 Observation

The simulator reports productivity, floor load, enacted support, public-benefit proxy, cooperation, compromise, gridlock, access denial, committee rejection, challenge rate, low-support passage, policy shift, proposer gain, lobby capture, public alignment, anti-lobbying success, private-gain ratio, lobbying spend by channel, public benefit per lobby dollar, amendment rate, amendment movement, minority harm, concentrated-harm passage, compensation rate, legitimacy, active-law welfare, reversal rate, correction delay, low-support active-law share, selected-alternative distance, proposer agenda advantage, alternative diversity, status-quo win rate, citizen-review metrics, attention spending, objection-window use, public-will review, district alignment, cosponsorship, bond forfeiture, strategic decoys, proposer-access Gini, welfare per submitted bill, vetoes, overrides, chamber conflict, malapportionment, population-seat distortion,

committee capture, eligibility exclusion, renewal staggering, recusal, ex ante review, review delay, and quiet-capture risk.

5 Initialization

`WorldGenerator` creates legislators from party-system and ideology parameters, derives party positions, generates lobby groups by issue domain, chooses a scalar status quo, and generates bills around proposer ideal points. Generated values are bounded by model validation. Party-system profiles include ideological bins, two major parties with minor parties, fragmented multiparty systems, dominant-party systems, and weighted campaign profiles.

6 Input Data and Calibration

Normal scenario runs use synthetic generated worlds. Calibration screening reads the tracked benchmark extract at `data/calibration/empirical-benchmarks.csv`. The extract maps empirical sources to simulator metrics and benchmark ranges. Current named sources include Voteview roll-call data [Lewis et al. 2021], Congress.gov and govinfo bill histories [Library of Congress 2026; United States Government Publishing Office 2020], Comparative Agendas Project topic data [Comparative Agendas Project 2026], ParlGov party-system data [Döring and Manow 2024], U.S. Lobbying Disclosure Act filings [United States Senate Office of Public Records 2026], Center for Effective Lawmaking scores [Center for Effective Lawmaking 2026; Volden and Wiseman 2014], and V-Dem institutional indicators [V-Dem Institute 2026].

The calibration runner executes conventional scenarios and writes CSV and Markdown pass/fail reports under `reports/`. The current calibration pass is an empirical screen, not an automatic parameter fitter. Its benchmark ranges are intentionally broad and several are abstract proxies, so passing the screen should be read as minimum plausibility rather than validation. Its purpose is to keep ordinary baselines within explicit plausible ranges before interpreting counterfactual mechanisms.

7 Submodels

7.1 Voting

Legislators cast `YAY` or `NAY` through weighted influences: ideology, party, constituency/public support, lobbying, compromise preference, reputation, and affected-group sensitivity.

7.2 Proposal Access and Agenda Scarcity

Access rules include open access, viability screens, scalar proposal costs, public-value waivers, lobby surcharges, member quotas, leadership agenda control, public-interest screens, cross-bloc cosponsorship gates, affected-group sponsor gates, proposal bonds, earned proposal credits, agenda lotteries, and quadratic attention budgets.

7.3 Committees and Information

Committee gatekeeping can reject proposals before floor consideration. Committee information processes move public-signal estimates toward generated public benefit with composition-dependent accuracy. Configured committee selection can vary issue domain, random seed, party quota, expertise weighting, independence weighting, leadership control, rotation memory, chair allocation, quorum party balance, citizen advisory weight, and topic-referral authority.

Committee-power processes distinguish priority queues, veto players, fast-track certification, scrutiny and audit, public accounts, legal/drafting review, and discharge-petition targets.

7.4 Chamber Structure, Eligibility, and Independent Review

Chamber specifications define equal-population, proportional, mixed-member, territorial, malapportioned, appointed, indirect, functional, or mixed elected/appointed chambers. Bicameral routing variants can change origination, emergency routing, second-chamber preclearance, amendment-only powers, absolute or territorial vetoes, suspensive vetoes, navette, mediation, last-offer bargaining, joint sittings, and lower-house overrides. Eligibility and retention filters model stylized age, residency, language, expertise, conflict, attendance, capture-risk, dual-office, party/executive overlap, recusal, cooling-off, term-limit, recall, public-support retention, and sanctions rules. Ex ante review variants model optional referral, mandatory high-risk review, advisory opinions, suspensive objections, mandatory clearance, qualified override, and deadline-limited review by insulated fiscal, electoral, audit, ethics, cost-estimator, regulator, or boundary institutions.

7.5 Challenge and Objection

Challenge vouchers allocate scarce blocking rights to parties or members. Q-member escalation lets a sufficient number of legislators trigger active review. Public objection windows route contested bills to active voting, higher thresholds, or repeal review depending on scenario.

7.6 Lobbying

Budgeted lobbying modifies bill pressure and public signals through channel-specific spending. Anti-capture variants include transparency, public financing, public advocates, blind review, audits, sanctions, and defensive-spend caps. Adaptive lobbying learns from channel returns and reform threats.

7.7 Amendment and Mediation

Structured mediation moves risky proposals toward weighted institutional targets. Multi-round mediation models repeated amendments, costs, concession limits, poison-pill risk, and compensation for concentrated harm.

7.8 Distributional Harm

Harm-weighted thresholds, compensation amendments, and affected-group consent mechanisms distinguish broad aggregate benefit from concentrated minority loss.

7.9 Citizen Mini-Publics

Citizen panels simulate sampling noise, information quality, manipulation risk, informed support, benefit estimates, and legitimacy. Panel output can affect default-pass eligibility, active-vote routing, final threshold, agenda priority, or affirmative majority burden of proof.

7.10 Leadership, Direct Democracy, and Judicial Review

Leadership agenda control schedules bills using observable public support, salience, uncertainty, majority-party fit, capture risk, and concentrated-harm risk rather than generated true welfare. Citizen-initiative processes let sufficiently salient, high-support proposals bypass legislative floor blockage through a public referendum. Conference-committee

processes revise near-miss bicameral bills toward a bargaining target before chamber revotes. Judicial-review processes can invalidate enacted laws with high rights-like harm, uncertainty, policy shock, or capture risk.

7.11 Alternatives, Package Bargaining, Norm Erosion, and Law Registry

Competing-alternative processes generate same-domain alternatives and select by public benefit, public support, chamber-median proximity, or pairwise majority before final ratification. Package-bargaining processes approximate compromise through side-payment costs, implementation-delay costs, modest policy movement, and harm-reducing package trades. Norm-erosion processes raise procedural delay and reduce public support signals under high contention. Law-registry processes allow enacted laws to persist, be reviewed later, and be renewed, repealed, or partially rolled back.

8 Experimental Design

The main submitted evidence comes from the main comparison campaign. That campaign joins broad assumption cases, adversarial proposal-generator cases, party-system sensitivity cases, and a stylized rising-contention timeline in one artifact. It is breadth-first: it includes the stylized U.S.-like benchmark, conventional affirmative baselines, leadership agenda control, committee-first regular order, cloture with conference and judicial review, parliamentary-style coalition confidence, direct citizen initiatives, non-default policy tournaments, citizen assembly review, public-interest screening, agenda lotteries, quadratic attention budgets, proposal bonds, harm-weighted thresholds, compensation amendments, package and omnibus bargaining, public objection, law-registry review, anti-capture and influence-system safeguards, district-public signals, adaptive and long-horizon risk routing, constitutional-court architecture, portfolio hybrids, norm erosion, and a small default-enactment stress-test family. Adversarial generator cases test high-benefit extreme reform, popular harmful bills, moderate capture, delayed-benefit reform, concentrated rights-like harm, and anti-lobbying backlash. Earlier development campaigns remain historical artifacts, including deeper default-pass sweeps, but are not mixed into the main paper evidence base. Follow-up diagnostics include mechanism ablations, manipulation-stress campaigns, chamber-structure family/stress screens, and an empirical bridge that maps optional raw empirical summaries to current simulator proxy metrics.

9 Assumptions and Limitations

The policy space is one-dimensional. Public benefit is generated rather than empirically estimated. Legislators are synthetic and do not represent named real officials. Lobby groups are abstract collective actors. The calibration layer screens plausible ranges but does not yet fit raw datasets automatically. Leadership, initiative, conference, judicial, court-architecture, district-public, influence-system, long-horizon learning, and norm-erosion processes are stylized mechanism prototypes rather than detailed institutional models. Institutional mechanisms are simplified so they can be compared in bundles without modeling the full legal, administrative, judicial, media, and electoral environment. Omnibus and package bargaining are proxies for side payments, delays, exemptions, jurisdictional trades, distributive load, and harm reduction, not full models of legal text, appropriations, or multidimensional coalition formation.

10 Reproducibility

The main reproducibility entrypoints are:

- `make test`: compile and run the simulator test suite.

Table 1. Full scenario averages from the main comparison campaign. Metric entries are mean±case half-range across broad and adversarial cases; Seed dir. reports directional-score mean±half-range across the independent seed sweep.

Label	Scenario	Directional ↑	Seed dir. ↑	Prod. ↑	Comp. ↑	Enacted/run	Weak mandate ↓	Admin cost ↓	Risk ctrl. ↑
CUR	**Stylized U.S.-like conventional benchmark**	**0.653±0.033**	**0.650±0.003**	**0.052±0.103**	**0.368±0.184**	**4.34±7.27**	**0.007±0.022**	**0.016±0.014**	**0.929±0.045**
SM	Unicameral simple majority	0.666±0.096	0.666±0.000	0.320±0.458	0.245±0.193	26.94±38.42	0.213±0.291	0.070±0.000	0.852±0.087
S60	Unicameral 60 percent passage	0.647±0.073	0.646±0.003	0.131±0.379	0.355±0.228	10.18±22.72	0.062±0.145	0.070±0.000	0.902±0.091
BIC	Bicameral simple majority	0.654±0.091	0.655±0.000	0.225±0.438	0.286±0.193	18.25±26.25	0.153±0.252	0.070±0.000	0.876±0.092
VETO	Bicameral majority + presidential veto	0.642±0.070	0.643±0.001	0.149±0.325	0.306±0.211	12.02±19.47	0.122±0.204	0.070±0.000	0.890±0.077
LEAD	Leadership agenda cartel + majority	0.657±0.054	0.657±0.001	0.162±0.180	0.269±0.185	14.29±24.95	0.117±0.199	0.027±0.014	0.906±0.042
COMM	Committee-first regular order	0.664±0.070	0.665±0.001	0.207±0.409	0.287±0.184	17.01±24.53	0.098±0.119	0.032±0.022	0.891±0.099
PROC	Cloture, conference, and judicial review	0.642±0.034	0.642±0.003	0.047±0.079	0.325±0.205	3.91±6.58	0.032±0.045	0.033±0.017	0.929±0.039
COURT	Constitutional court architecture	0.652±0.082	0.653±0.001	0.320±0.461	0.246±0.192	26.93±38.42	0.213±0.293	0.108±0.055	0.834±0.082
PARL	Parliamentary coalition confidence	0.672±0.144	0.672±0.000	0.225±0.425	0.292±0.267	18.41±27.57	0.067±0.194	0.019±0.030	0.904±0.115
INIT	Citizen initiative and referendum	0.693±0.079	0.694±0.000	0.430±0.385	0.221±0.209	37.76±50.50	0.223±0.278	0.070±0.000	0.833±0.072
DIST	District-population public-will model	0.654±0.068	0.654±0.001	0.268±0.430	0.261±0.170	22.44±31.48	0.347±0.369	0.070±0.000	0.856±0.133
PAIR	Unicameral majority + pairwise alternatives	0.712±0.039	0.712±0.001	0.542±0.207	0.502±0.093	48.78±68.64	0.002±0.003	0.285±0.044	0.937±0.067
JURY	Citizen assembly threshold gate	0.631±0.053	0.631±0.001	0.197±0.175	0.283±0.216	17.12±25.76	0.061±0.227	0.190±0.000	0.907±0.077
SCR	Majority + public-interest screen	0.657±0.062	0.657±0.001	0.246±0.226	0.248±0.191	21.66±36.03	0.183±0.295	0.060±0.030	0.872±0.053
LOT	Weighted agenda lottery + majority	0.646±0.045	0.647±0.000	0.173±0.217	0.250±0.194	14.84±22.74	0.183±0.279	0.033±0.001	0.877±0.052
QAB	Quadratic attention budget + majority	0.620±0.071	0.620±0.000	0.247±0.366	0.265±0.189	19.94±22.98	0.178±0.292	0.220±0.033	0.878±0.076
BOND	Proposal bonds + majority	0.666±0.096	0.666±0.000	0.319±0.455	0.245±0.195	26.76±37.65	0.213±0.290	0.070±0.002	0.852±0.084
HARM	Harm-weighted double majority	0.662±0.098	0.663±0.001	0.271±0.449	0.261±0.196	23.45±36.33	0.202±0.286	0.070±0.000	0.876±0.069
COMP	Compensation amendments + majority	0.662±0.097	0.662±0.001	0.320±0.460	0.245±0.197	26.84±38.10	0.214±0.289	0.070±0.000	0.835±0.077
PKG	Package bargaining + majority	0.655±0.103	0.656±0.000	0.329±0.450	0.243±0.199	27.62±38.56	0.223±0.293	0.110±0.040	0.841±0.082
MPKG	Multidimensional package bargaining + majority	0.640±0.115	0.641±0.001	0.341±0.443	0.240±0.199	28.61±39.49	0.218±0.205	0.173±0.116	0.835±0.081
OMNI	Omnibus bargaining + majority	0.629±0.116	0.630±0.001	0.376±0.420	0.236±0.199	31.53±41.51	0.157±0.149	0.262±0.127	0.846±0.082
LAW	Active-law registry + majority review	0.654±0.082	0.655±0.001	0.346±0.463	0.236±0.175	29.34±41.91	0.244±0.301	0.118±0.048	0.835±0.090
OBJ	Public objection window + majority	0.644±0.082	0.644±0.000	0.229±0.379	0.283±0.199	19.30±29.65	0.123±0.263	0.134±0.017	0.892±0.074
CAP	Majority + anti-capture bundle	0.675±0.056	0.676±0.001	0.306±0.197	0.247±0.177	26.86±39.16	0.152±0.226	0.065±0.016	0.884±0.056
INFL	Campaign-finance influence system	0.674±0.060	0.675±0.001	0.316±0.216	0.244±0.193	27.60±41.26	0.142±0.195	0.062±0.022	0.871±0.060
RISK	Risk-routed majority legislature	0.687±0.073	0.687±0.001	0.530±0.331	0.264±0.170	45.41±60.13	0.231±0.171	0.171±0.029	0.843±0.070
PORT	Portfolio hybrid legislature	0.705±0.068	0.706±0.000	0.503±0.303	0.421±0.063	45.39±71.56	0.030±0.026	0.240±0.096	0.933±0.034
XPORT	Expanded portfolio hybrid legislature	0.676±0.074	0.676±0.001	0.784±0.239	0.463±0.052	70.01±102.46	0.002±0.004	0.607±0.107	0.905±0.052
NORM	Endogenous norm erosion + majority	0.631±0.091	0.632±0.001	0.325±0.426	0.239±0.203	27.46±38.78	0.251±0.301	0.200±0.005	0.843±0.087
LEARN	Long-horizon strategic learning	0.679±0.071	0.679±0.001	0.591±0.296	0.269±0.164	51.61±69.53	0.242±0.175	0.210±0.001	0.798±0.044
DP	Default pass unless 2/3 block	0.725±0.049	0.725±0.001	0.866±0.089	0.136±0.221	79.23±101.73	0.472±0.370	0.070±0.000	0.630±0.065
DPC	Default pass + challenge vouchers	0.683±0.087	0.683±0.002	0.659±0.321	0.161±0.217	67.83±127.35	0.450±0.355	0.105±0.028	0.693±0.123
DPM	Default pass + multi-round mediation + challenge	0.707±0.056	0.708±0.002	0.802±0.177	0.243±0.171	77.56±121.99	0.374±0.226	0.206±0.042	0.722±0.070

Note: Case half-ranges are descriptive variation across campaign cases, not statistical confidence intervals. Directional, productivity, compromise, weak public-mandate passage, administrative cost, and risk-control values are normalized scores or rates. *Enacted/run* is an absolute institutional-load count.

- `make campaign` or `make paper-campaign`: regenerate the current comparison campaign.
- `make calibrate`: run empirical benchmark screening.
- `make seed-robustness-check`: run multi-seed checks for paper scenarios.
- `make validation-readiness`, `make empirical-validation`, and `make empirical-bridge`: check optional raw inputs, summarize any present data, and map summaries to simulator proxies.
- `make ablation-analysis`, `make manipulation-stress`, and `make mechanism-diagnostics`: run mechanism ablations, manipulation stress tests, and the generated diagnostics table workflow.
- `make family-screen`, `make catalog-breadth`, and `make chamber-structure`: run breadth-catalog and chamber-structure diagnostics.
- `make paper`: build the paper and appendix PDFs.
- `make supplement-anonymous`: build the double-blind supplement package.
- `make public-provenance`: after review, generate a non-anonymous provenance manifest.

The review package includes generated campaign reports and PDFs. Public repository identifying details are withheld during double-blind review and can be restored after review if permitted.

Use of Large Language Model Tools

The research question and core simulator idea originated with the human author. Large language model tools were used for software implementation assistance and for drafting, editing, and reorganizing manuscript text. The author

reviewed and is responsible for all code, experiments, reported results, text, and citations. No large language model is listed as an author.

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